

PORTFOLIO PROJECTS

PROJECT 1: Enterprise System Administration Infrastructure

ORGANIZATION: AtronClean Industrial Services Ltd.

TECHNOLOGIES;

- ❖ Ubuntu Server 22.04 LTS
- ❖ Windows Server
- ❖ Active Directory
- ❖ Samba
- ❖ DNS
- ❖ DHCP
- ❖ SSH
- ❖ File Server
- ❖ Backup Solutions

Achievements

- ❖ Configured centralized file services
- ❖ Implemented user and group management
- ❖ Created secure access permissions
- ❖ Designed automated backup procedures
- ❖ Improved system security through access control
- ❖ Implemented disaster recovery planning

Reflection

This project strengthened my understanding that effective system administration is more than maintaining servers; it is about ensuring users can work securely, efficiently, and reliably while safeguarding organizational resources.

PROJECT 2: Enterprise Linux File Server Deployment

SKILLS DEMONSTRATED;

- ❖ Linux administration
- ❖ Shell scripting
- ❖ User account management
- ❖ File permissions
- ❖ Group policies
- ❖ Backup automation
- ❖ Network configuration

Reflection

Managing enterprise servers reinforced the importance of accountability, careful planning, and maintaining system availability for organizational success.

PROJECT 3: IT Infrastructure Planning and Security Assessment

INCLUDED;

- ❖ Network assessment
- ❖ Risk analysis
- ❖ Security hardening
- ❖ Infrastructure documentation
- ❖ Backup strategy
- ❖ Disaster recovery planning
- ❖ System maintenance procedures

Reflection

This project demonstrated that successful infrastructure management requires balancing technical efficiency with ethical responsibility, particularly regarding data protection, user privacy, and business continuity.

ETHICAL DILEMMA

Original Situation

The organization planned to give all employees local administrator privileges on their workstations to reduce help desk requests and simplify software installation.

Essential Ethical Principles

- ❖ Responsibility
- ❖ Integrity
- ❖ Accountability
- ❖ Security
- ❖ Fairness
- ❖ Professional competence

Alternative Solution

Instead of granting administrator privileges to every employee, implement Role-Based Access Control (RBAC) following the Principle of Least Privilege. Standard users should receive only the permissions necessary for their roles, while administrative access is restricted to authorized IT personnel. When elevated privileges are required, use temporary privilege elevation with proper approval, auditing, and documentation.

Why This Alternative Is Better

This approach significantly reduces the risk of malware infections, accidental system changes, unauthorized software installations, and data breaches. It also strengthens accountability by ensuring that administrative actions are logged and traceable. While users may occasionally wait for IT assistance, the organization benefits from greater security, regulatory compliance, and system stability.